System Architecture & Integration

Multi-Tier Design, Micro-Router Breakdown, PostgreSQL Schema & Security Topology

Document Title: System Architecture & Integration Specification
Architecture Style: Modular Micro-Router REST API + High-Performance Web Client
Backend Framework: FastAPI 0.115+ (Asynchronous ASGI) with Pydantic v2
Database & ORM: PostgreSQL (Production) with SQLite Automatic Failover
Security Standard: OAuth2 Password Bearer Flow with Role-Based Access Control

1. Architectural Overview & Component Decomposition

The system is architected as a modular, decoupled platform where the frontend client, REST API layer, ML scoring engine, and database communicate via strictly typed interfaces:

Module / Component	File Location	Core Architectural Function
FastAPI Entry Point	backend/app.py	Slim ASGI gateway (<50 lines); mounts CORS, lifespan, and router endpoints
Application Config	backend/config.py	Centralized environment settings, CORS origins, model paths, and logging
Authentication Service	backend/auth.py	OAuth2 scheme, JWT / Bearer token validation, password verification, RBAC
Prediction Router	backend/routes/prediction_routes.py	Handles /predict, /batch-predict, and /sample-197-batch endpoints
Alert Router	backend/routes/alert_routes.py	Handles /alerts, /status, /alerts-summary CRUD operations
System Router	backend/routes/system_routes.py	Handles /health and /model/monitoring-stats endpoints
Model Service	backend/model_service.py	In-memory model singleton, feature alignment, calibrated scoring (T=0.42)
Alert Service	backend/alert_service.py	SQLAlchemy PostgreSQL engine, connection pooling, SQLite failover

2. PostgreSQL Database Schema (`fraud_alerts`)

Alerts generated from suspicious predictions ($P \ge 0.42$) are persisted in PostgreSQL. The schema ensures idempotency, auditability, and fast querying:

Column Name	Data Type	Constraints	Description
id	BIGINT	PRIMARY KEY, AUTOINCREMENT	Unique internal database record identifier
transaction_id	VARCHAR(255)	UNIQUE, NOT NULL, INDEXED	Banking transaction identifier (idempotent upsert key)
fraud_probability	FLOAT	NOT NULL	Calibrated probability score output by ML model
prediction	VARCHAR(50)	NOT NULL	Human-readable verdict ('Suspicious' / 'Normal')
risk_score	FLOAT	NOT NULL	Percentage risk rating (0.0 to 100.0)
risk_level	VARCHAR(20)	NOT NULL	Assigned risk tier ('Low', 'Medium', 'High')
status	VARCHAR(20)	NOT NULL, DEFAULT 'New'	Triage workflow state ('New', 'Under Review', 'Resolved')

created_at	TIMESTAMP WITH TZ	NOT NULL, DEFAULT NOW()	Timestamp when alert was first ingested
updated_at	TIMESTAMP WITH TZ	NOT NULL, DEFAULT NOW()	Timestamp of most recent status modification

3. Security & Role-Based Access Control (RBAC)

All sensitive endpoints require an HTTP Authorization header containing a valid Bearer token. Two distinct enterprise roles are supported:

- **Administrator / Chief Risk Officer (admin):** Full administrative privileges including prediction, status transitions, batch scoring, and alert record deletion.
- **Fraud Analyst (analyst):** Triage permissions for single/batch scoring and updating alert statuses from 'New' to 'Under Review' or 'Resolved'.

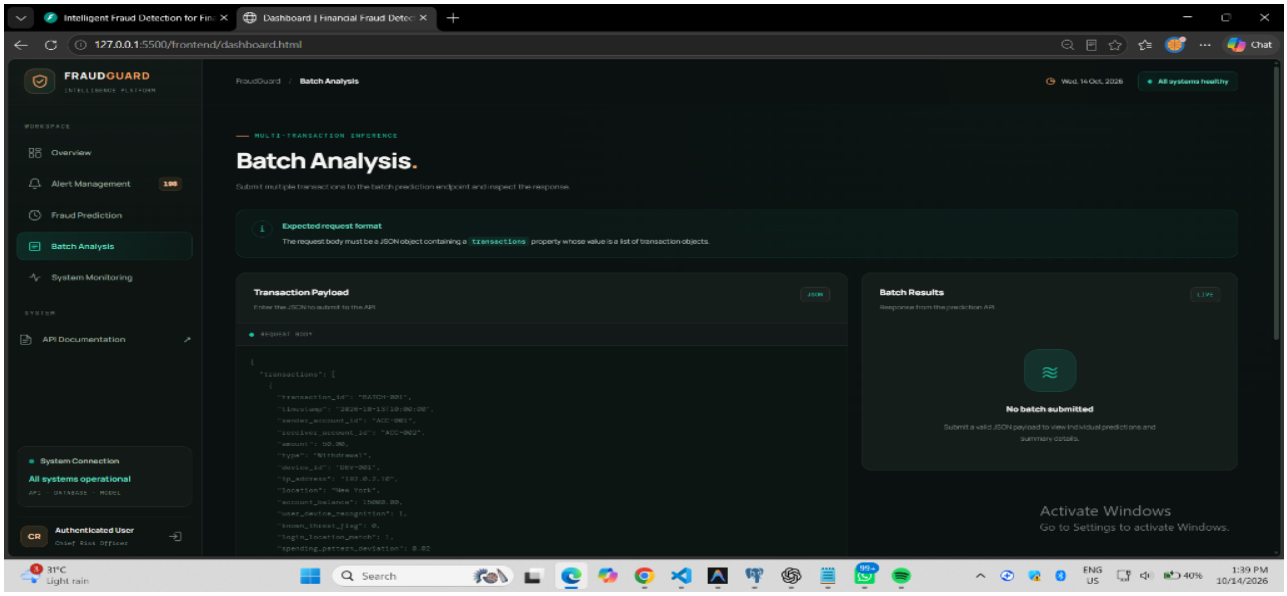


Figure 1: Secure Authentication Portal with Password Toggle and Theme Synchronization